# DBA5103 • OPERATIONS RESEARCH & ANALYTICS

**SIDE 1 / 4 · LP, SIMPLEX, DUALITY & UNCERTAINTY**

## 1 · LP FOUNDATIONS & MODELLING [L1–2]

**Workflow:** define variables + units → objective → every constraint → variable domains → check feasibility and units. LP is linear in **decision variables**; coefficients may be nonlinear functions of known data. E.g. $a^2x + \sqrt{a}\,y \le 1$ is linear when a is a fixed parameter; xy and $x^2$ are not linear.

**LP assumptions:** proportionality, additivity, divisibility (fractions allowed), known coefficients. LP = continuous; IP = discrete; MIP = both; DP = sequential decisions with a sufficient state. Revenue p·x is linear only if p is fixed.

**Standard form used in class:** min $c^Tx$, $Ax = b$, $x \ge 0$. Max $c^Tx = -\min(-c^Tx)$. Row "≤": add slack +s; "≥": subtract surplus −s; $s \ge 0$. Free $x = x^+ - x^-$; $x^+, x^- \ge 0$. If $x \le 0$, substitute $x = -u$, $u \ge 0$. Lower bound $x \ge \ell$: $x = \ell + u$.

**Example:** min $2x + 3y$, $x - y \ge 4$, x free, $y \le 0$ becomes min $2x^+ - 2x^- - 3u$; $x^+ - x^- + u - s = 4$; all new variables $\ge 0$.

**Production/procurement:** quantities x, purchases y; max $p^Tx - c^Ty$; $Ax \le y$, $c^Ty \le B$, $x,y \ge 0$. With nonnegative costs and no other purchase restrictions, eliminate y: max $(p^T - c^TA)x$; $c^TAx \le B$.

**Diet / staffing:** min $c^Tx$; nutrient bounds $L \le Ax \le U$. For shifts, $A_tj = 1$ if shift j covers period t; $\sum_j A_tj\, x_j \ge$ demand_t. Pay **whole-shift** cost. Full-time ≥ part-time must hold separately each period.

**Absolute values:** $|r| \le t \Leftrightarrow r \le t$ and $-r \le t$. Min $\sum w_i|r_i|$ ($w_i \ge 0$): min $\sum w_i\, t_i$, $t_i \ge \pm r_i$. Min $\max_i|r_i|$: min t, $t \ge \pm r_i\ \forall i$. $\sum|x_i| \le B$: $t_i \ge \pm x_i$, $\sum t_i \le B$. This trick does **not** enforce $t = |r|$ if objective/constraints do not push t down; $|x| \ge b$ is generally nonconvex.

**Example:** min $2x_1 + 3|x_2 - 10|$, $|x_1+2| + |x_2| \le 5 \rightarrow$ min $2x_1+3t$; $t \ge x_2 - 10$, $t \ge 10 - x_2$; $u \ge \pm(x_1+2)$, $v \ge \pm x_2$; $u+v \le 5$. Keep $x_1, x_2$ free.

**Regression/sparsity:** residual $r_i = b_i - a_i^T\beta$. L1 = sum absolute residuals (LP, less outlier-sensitive); L∞ = largest absolute residual (LP); L2 = sum squared residuals (quadratic). Sparse recovery: min $\sum|x_j|$ s.t. $Ax = b$. For $x_1+2x_2=2$: L1 gives (0,1); L2 gives (0.4,0.8).

## 2 · GEOMETRY, BASIS & SIMPLEX [L2–3]

**Linear algebra:** A is m×n, Ax=b; $(AB)^T = B^TA^T$. Square A invertible ⇔ independent columns/rows ⇔ $\det A \ne 0$. For A=[a b; c d], $A^{-1}$=[d −b; −c a]/(ad−bc). rank = number of independent columns = independent rows. Nonzero outer product $uv^T$ has rank 1.

**Convex:** $\lambda x+(1-\lambda)y$ is feasible for $\lambda \in [0,1]$. Convex f: $f(\lambda x+(1-\lambda)y) \le \lambda f(x)+(1-\lambda)f(y)$. Convex minimization: every local minimum is global. LP is convex; IP generally is not. Polyhedron = finite intersection of linear halfspaces/hyperplanes.

**BFS/extreme point:** n independent active constraints in n variables (include nonnegativity). For Ax=b, rank A=m: choose m independent columns B; set x_N=0; $x_B = B^{-1}b$. Feasible iff $x_B \ge 0$. At most m positive variables; a basic variable may be zero (**degenerate**). ≤ C(n,m) bases. Different bases may represent the same point.

**Examples:** $x_1+x_2+x_3=1$, $x \ge 0$ has BFS (1,0,0),(0,1,0), (0,0,1); (½,½,0) is not BFS. $[0,1]^3$ has 8 vertices. Counting active rows alone fails if rows are dependent.

**LP outcomes:** finite optimum (unique or multiple), infeasible, unbounded objective. Unbounded feasible set need not mean unbounded optimum. A feasible standard-form LP with finite optimum has an optimal BFS. General polyhedra with lines need not have vertices.

**Simplex, MIN convention:** from a BFS choose entering nonbasic j. Set $d_j=1$, other nonbasic d=0, $d_B = -B^{-1}A_j$. Reduced cost $r_j = c_j - c_B^TB^{-1}A_j = c^Td$. If all $r_j \ge 0$, optimal. Otherwise enter $r_j < 0$; $\theta^* = \min_{i:d_Bi<0}$ x_Bi/(−d_Bi). Update $x \leftarrow x + \theta^* d$; minimizing-ratio basic variable leaves. No negative d_B entry ⇒ unbounded. For MAX, enter $r_j > 0$; optimal when all $r_j \le 0$.

**Worked pivots:** max $x_1+x_2$; $x_1+2x_2+s_1=3$, $2x_1+x_2+s_2=3$. Start (0,0,3,3). Enter $x_2$: d=(0,1,−2,−1), θ=min(3/2,3)=1.5 → (0,1.5,0,1.5). Enter $x_1$: d=(1,−½,0,−1.5), θ=min(3,1)=1 → (1,1,0,0), z=2. For min $-x_1-x_2$, final slack reduced costs =⅓,⅓ ≥0.

**Pivot traps:** ratio test uses only decreasing basic variables. θ=0 means degeneracy, not infeasibility. Zero reduced cost permits another optimum if a positive feasible step exists. Adjacent bases exchange one column.

**Gurobi essentials (lecture code):** from gurobipy import *; m=Model(); x=m.addVars(n) (default x≥0, continuous); m.setObjective(quicksum(c[j]*x[j] for j in range(n)), GRB.MAXIMIZE); m.addConstr(...); m.optimize(). Solution: x[j].x, m.objVal. Use vtype=GRB.BINARY or GRB.INTEGER when needed; lb=-GRB.INFINITY for free variables. Check solver status before reading a solution.

## 3 · DUALITY & OPTIMALITY [L4]

**Two main pairs (A transposes):**

MAX $c^Tx$; $Ax \le b$, $x \ge 0$ ↔ MIN $b^Ty$; $A^Ty \ge c$, $y \ge 0$.

MIN $c^Tx$; $Ax \ge b$, $x \ge 0$ ↔ MAX $b^Ty$; $A^Ty \le c$, $y \ge 0$.

**Standard-form pair:** min $c^Tx$, $Ax=b$, $x \ge 0$ ↔ max $b^Ty$, $A^Ty \le c$, y free. Primal rows ↔ dual variables; primal variables ↔ dual rows. A is m×n ⇒ dual has m variables and n constraints.

**Mixed-sign recipe, MAX primal:** primal row ≤ / ≥ / = gives y ≥0 / ≤0 / free. Primal variable $x_j \ge 0$ / ≤0 / free gives dual row $(A^Ty)_j \ge c_j$ / $\le c_j$ / $= c_j$. For a **MIN primal**, reverse these inequality directions and y signs; equality/free stays unchanged.

**Mixed example:** max $3x_1+2x_2$; $x_1+x_2 \le 4$, $2x_1-x_2=1$; $x_1 \ge 0$, $x_2$ free. Dual: min $4y_1+y_2$; $y_1+2y_2 \ge 3$, $y_1-y_2=2$; $y_1 \ge 0$, $y_2$ free.

**Weak duality:** any feasible MAX-primal x and MIN-dual y satisfy $c^Tx \le b^Ty$. Thus a feasible dual is an upper bound and feasible primal a lower bound. Equal objectives certify both optimal. **Strong duality:** finite LP optimum ⇒ dual optimum exists with equal value. Unbounded primal ⇒ infeasible dual. Infeasible primal ⇒ dual may be infeasible OR unbounded; no converse shortcut.

**Complementary slackness, max/≤ pair:**

$y_i[b_i - (Ax)_i] = 0\ \forall i$; $x_j[(A^Ty)_j - c_j] = 0\ \forall j$.

Together with primal + dual feasibility these certify optimality. Positive primal slack ⇒ $y_i=0$; $y_i>0$ ⇒ row binding. $x_j>0$ ⇒ dual row j tight. Binding does **not** imply positive y; y=0 does not imply slack.

**Solve dual from known primal:** identify slack primal rows → set corresponding y=0; for positive $x_j$ solve $A_j^Ty = c_j$; then check all signs, remaining dual inequalities and equal objective.

**Wyndor anchor:** max $3x_1+5x_2$; $x_1 \le 4$, $2x_2 \le 12$, $3x_1+2x_2 \le 18$, $x \ge 0$. Vertices/objectives: (0,0):0; (4,0):12; (4,3):27; (2,6):36; (0,6):30. Optimum x*=(2,6), slacks=(2,0,0).

**Wyndor dual:** min $4y_1+12y_2+18y_3$; $y_1+3y_3 \ge 3$, $2y_2+2y_3 \ge 5$, $y \ge 0$. Slack row 1 ⇒ $y_1=0$; $x_1, x_2 > 0 \Rightarrow 3y_3=3$, $2y_2+2y_3=5$. Hence y*=(0,1.5,1); dual value 36 = primal value.

**Economic meaning:** $y_i$ = marginal objective value of one more unit of resource i, locally. $A_j^Ty$ = opportunity cost of resources for product j. At optimum, produced products earn exactly this resource value; unproduced products cannot exceed it at an optimal dual price.

**Lagrangian:** for min $c^Tx$, $Ax=b$, $x \ge 0$, $L = c^Tx + y^T(b - Ax)$. $g(y) = \inf_{x \ge 0} L$ is $b^Ty$ if $A^Ty \le c$, otherwise $-\infty$. Maximize g to get strongest lower bound. For nonlinear problems strong duality need not hold.

**Example:** min $x^2+y^2$, $x+y=1$. $L = x^2+y^2+p(1-x-y)$; minimizing gives $x=y=p/2$, $g(p) = p - p^2/2$. Max at p=1: g=½, x=y=½. Any p gives a lower bound.

## 4 · SENSITIVITY / WHAT-IF [L4]

**With same optimal basis B:** $x_B = B^{-1}b$; dual $y^T = c_B^TB^{-1}$; $r_N = c_N - A_N^Ty$. RHS changes: $x_B(\text{new}) = B^{-1}(b+\Delta b)$; maintain $x_B(\text{new}) \ge 0$. Objective changes: recompute y and r_N; MIN needs $r_N \ge 0$ (MAX ≤0).

**RHS value:** $z(\text{new}) = z^* + y^T\Delta b$ while basis stays optimal.

**Objective-coefficient change:** $z(\text{new}) = z^* + \Delta c^Tx^*$ while x* remains optimal. Sensitivity ranges are normally **one change at a time**; simultaneous changes require joint basis checks. A boundary can create degeneracy/alternative optima.

**Wyndor resource 3:** $b_3 = 18+\delta \Rightarrow x_2=6$, $x_1=2+\delta/3$, $s_1=2-\delta/3$. Feasible for $-6 \le \delta \le 6$; $z = 36+\delta$. +3 hours ⇒ z=39. +9 hours exceeds range: optimum (4,6), z=42, **not 45**.

**Other RHS ranges:** varying $b_1$ alone: $b_1 \ge 2$ (no upper limit), z=36. Varying $b_2$ alone: $6 \le b_2 \le 18$, $x_2 = b_2/2$, $x_1 = (18-b_2)/3$, $z = 18+1.5b_2$.

**Wyndor profit change:** hold $c_2=5$. $y_3 = c_1/3$, $y_2 = 2.5 - c_1/3$; $y \ge 0$ iff $0 \le c_1 \le 7.5$. Then (2,6) stays optimal, $z = 2c_1+30$. $c_1=4 \Rightarrow z=38$; $c_1=10 \Rightarrow$ new optimum (4,3), z=55. Hold $c_1=3$: same basis for $c_2 \ge 2$.

**New constraint (MAX):** optimum cannot rise. If old optimum satisfies it, value is unchanged. Example $x_1+x_2 \le 10$: (2,6) still feasible ⇒ z=36. A nonbinding constraint need not be globally redundant.

**New nonnegative variable (MAX):** optimum cannot fall because new variable can be 0. Compute $r_{new} = c_{new} - y^Ta_{new}$. If $r_{new} \le 0$, old dual stays feasible ⇒ old optimum remains. If >0, old dual fails; reoptimize (degeneracy may prevent strict improvement). Wyndor a=(2,4,6), c=6 ⇒ r=6−12=−6: no gain.

**Buying capacity:** compare extra profit with extra cost **within range**. Wyndor plant 3 offers 3 hours at 0.7 per hour: net gain=3(1−0.7)=0.9 (objective units). Shadow prices are not valid indefinitely.

**Bound intuition:** adding constraints shrinks feasible set: MAX ↓ / MIN ↑; relaxing constraints or adding optional variables: MAX ↑ / MIN ↓. For MAX, old x* gives a lower bound after a coefficient change; old feasible dual gives an upper bound after an RHS change.

## 5 · LP CASES & UNCERTAINTY [L5]

**Markdown LP:** price levels $p_k$, demand/week $d_k$, effective weeks $t_k \ge 0$, initial stock Q, horizon H, salvage s. Max $sQ + \sum_k (p_k - s)d_k\, t_k$; $\sum d_k\, t_k \le Q$; $\sum t_k \le H$; $t_{full} \ge 1$. Purchase cost is sunk. Order used levels high→low for no price increase. Fractional weeks allowed; model assumes time-independent demand rates.

**Lecture example:** Q=2000, H=15, s=25; (p,d)=(60,125), (54,162.5),(48,217.5),(36,348.8). Optimum t=(11⅔,3⅓,0,0). Check: 125(11⅔)+162.5(3⅓)=2000; revenue=60(1458⅓) +54(541⅔)=116750. At least 3 full-price weeks changes nothing. After poor sales: update **remaining** stock/time and solve again; past sales are fixed.

**Sensitivity vs uncertainty:** sensitivity changes known data and resolves; uncertainty chooses some actions **before** unknown data is observed. Nominal = optimize forecast/mean; robust = feasible for every realization in a specified set; stochastic = optimize expected outcome with probabilities/recourse.

**Drug sourcing case:** output x, raw purchases $q_1, q_2 \ge 0$. Max $5500x - 100q_1 - 199.9q_2$; $90x \le 2000$, $40x \le 800$, $q_1+q_2 \le 1000$, $700x + 100q_1 + 199.9q_2 \le 100000$. Ingredient: $.01q_1 + .02q_2 \ge .5x$.

**Box robustness:** raw content $a_1 \in [.00995, .01005]$, $a_2 \in [.0196, .0204]$. Since $q \ge 0$, robust ingredient row uses **lowest** yields: $.00995q_1 + .0196q_2 \ge .5x$. Nominal plan ≈$(x, q_1, q_2)$=(17.55,0,438.79), profit 8819.66; may violate realized yield. Robust plan ≈(17.47,877.73,0), profit 8294.57. Worst-case lower yield can change preferred supplier.

**Worst-case direction:** for $\sum a_j x_j \le b$ with $x \ge 0$, choose largest a and smallest b; for ≥ choose smallest a and largest b. With signed x use bounds/absolute values; cannot blindly pick one endpoint.

**Budgeted uncertainty:** $a_j = \hat{a}_j + \hat{a}_j\, z_j$, $|z_j| \le 1$, $\sum|z_j| \le \Gamma$. Lecture $\Gamma = 2\alpha$, $0 \le \alpha \le 1$: α=0 nominal; α=1 full box. Larger Γ ⇒ more protection, smaller robust feasible set.

**Compact robust LP (q≥0):** require $\min_z \sum(\hat{a}_j + \hat{a}_j\, z_j)q_j \ge R$. Equivalent: $\sum \hat{a}_j q_j - \Gamma\rho - \sum s_j \ge R$; $\rho + s_j \ge \hat{a}_j q_j$; $\rho, s_j \ge 0$. This uses duality for the worst-case loss $\max \sum \hat{a}_j q_j\, u_j$, $\sum u_j \le \Gamma$, $0 \le u_j \le 1$. Drug: $\hat{a}$=(.01,.02), $\hat{a}$=(.00005,.0004), R=.5x.

**Small example:** q=(100,100), nominal ingredient=3; losses=(.005,.04). Γ=0 ⇒ 3; Γ=1 ⇒ 3−max(.005,.04)=2.96; Γ=2 ⇒ 2.955. For fractional Γ, sort losses and take full largest losses plus the fraction of the next.

## 6 · SCENARIOS & RECOURSE [L6–7]

**Two-stage flexibility:** build links $y_{ij} \in \{0,1\}$ before demand; allocate $x_{ij}(D)$ afterward. Max $-\sum K_{ij}\, y_{ij} + E[Q(y,D)]$; $Q = \max \sum p_j\, x_{ij}$, with $\sum_i x_{ij} \le C_i$, $\sum_i x_{ij} \le D_j$, $0 \le x_{ij} \le M_{ij}\, y_{ij}$. Use a valid tight M, e.g. $C_i$.

**Sample average approximation (SAA):** K sampled demands D^k; one common y, separate x^k. Max $-\sum K_{ij}\, y_{ij} + (1/K)\sum_{k,ij} p_j x_{ij}^k$, with capacity, demand and link constraints **for each k**. For unequal probabilities use $\sum p_k Q_k$. Do not average demands first; E[min(C,D)] ≠ min(C,E[D]). Do not give first-stage y a scenario index: that assumes future knowledge.

**Mini-example:** capacity 1; D=0 or 2, each with probability ½. Expected optimal sales E[min(1,D)]=½. Using average demand E[D]=1 predicts sales 1 and overstates performance.

**Fixed-design simulation:** sample D, solve allocation LP, average optimal sales/profit. Lecture $D_j = \max(X_j, 0)$, $X_j$ normal: clipping at zero is not conditional resampling. Use the same demand samples to compare designs; increase sample size until the difference stabilizes.

**Inventory/transshipment:** first-stage stock $q_i$, per-scenario shipments $f_{ij}^k \ge 0$ (include i=j for local sales). $\sum_j f_{ij}^k \le q_i$; $\sum_i f_{ij}^k \le D_j^k$. Max $-c\sum q_i + (1/K)\sum_{k,ij}(p - t_{ij}) f_{ij}^k$. Subtract a service fixed cost once if paid regardless of use; an already paid cost does not affect future decisions.

## QUICK MODELLING / EXAM CHECKS

**Recognize patterns:** "at least" ≥; "at most" ≤; "exactly" =. Capacity constrains **total usage**; demand constrains **total delivery**. Index each constraint by what it protects. Percentages: 0.005%=0.00005, not 0.005.

**Profit vs revenue:** subtract variable purchase/production costs; ignore sunk costs when comparing future actions. "No sale" is feasible only if minimum demand/commitment constraints allow it. Add variable upper bounds when physically required.

**Portfolio example:** max $\sum r_i x_i$; $\sum x_i \le B$, $x \ge 0$. Put all budget in a highest positive-return stock; if all $r_i \le 0$, investing nothing is optimal. Dual min By, $y \ge r_i$, $y \ge 0 \Rightarrow y^* = \max(0, \max r_i)$. Tied best stocks permit split optima.

**Useful checks:** plug your answer into every row; calculate slacks; check signs and units; prove optimum using primal=dual value, reduced costs, a matching bound, or a Bellman comparison. Feasible ≠ optimal; binding ≠ valuable; an LP fractional optimum need not round to an IP optimum.

**Lecture map:** L1 OR/model choice; L2 LP + reformulation; L3 BFS/simplex; L4 duality/sensitivity; L5 markdown/robustness; L6 networks; L7 IP/relaxation/B&B; L8 DP. Examples marked "lecture" use course data; other small examples are added for practice.

# DBA5103 • OPERATIONS RESEARCH & ANALYTICS

**SIDE 2 / 4 · NETWORKS, INTEGER MODELS & DYNAMIC PROGRAMMING**

## 7 · NETWORK FLOW [L6]

**Graph facts:** G=(N,A); directed arcs are ordered. Tree = connected, acyclic undirected graph; spanning tree uses all n nodes and n−1 edges. Complete undirected graph: n(n−1)/2 edges; directed (no loops): n(n−1). Bipartite: edges only between two sets; complete bipartite m×n edges. Every tree is bipartite; not conversely.

**Min-cost flow:** min $\sum_{(i,j)} c_{ij} f_{ij}$; $\sum_j f_{ij}-\sum_j f_{ji}=b_i\ \forall i$; $\ell_{ij}\le f_{ij}\le u_{ij}$. **Out−in=b:** supply b>0, demand b<0, transit b=0; $\sum b_i=0$ necessary (not sufficient). Arc column has +1 at origin, −1 at destination. One balance row is redundant in a connected network; incidence rank n−1.

**Lower bounds:** set $f=\ell+g$; new capacity $u-\ell$; new supply $b'_i=b_i-\sum_j\ell_{ij}+\sum_j\ell_{ji}$. A required shipment d can be fixed by $\ell=u=d$. Prohibited arcs: omit or set u=0.

**Transportation:** min$\sum c_{ij}x_{ij}$; $\sum_j x_{ij}=a_i$ (supplies), $\sum_i x_{ij}=d_j$ (demands), x≥0. Balanced iff $\sum a=\sum d$; add a dummy for legitimate unused supply/unmet demand, using appropriate cost/penalty. Connected m+n-node transportation basis: m+n−1 tree arcs (some may have zero flow).

**Tree example (assignment):** supplies a,100−a; demands 70,30; set $x_{11}$=0. Balances give $x_{12}=a$, $x_{21}=70$, $x_{22}=30-a$. Feasible iff 0≤a≤30. At a=30, tree variable $x_{22}$=0: degenerate BFS. A spanning tree gives a basic solution; check nonnegativity for BFS.

**Initial transport solution:** northwest-corner allocates min(remaining supply,demand), removes exhausted row/column; least-cost method uses cheapest available cell. Both can need zero basic arcs in ties. Neither guarantees optimality.

**Transport pivot:** for basic arcs solve $u_i+v_j=c_{ij}$ (fix one potential=0). Nonbasic reduced cost $r_{ij}=c_{ij}-u_i-v_j$; MIN optimal if all r≥0. Enter a negative-r arc, form unique tree cycle, alternate +/−, send θ=min flow on "−" arcs; one such arc leaves.

**2×2 pivot:** costs [2,5;4,1], supplies (3,2), demands (2,3). Start $x_{12}$=3,$x_{21}$=2 (cost 23), add zero basic $x_{22}$. Enter $x_{11}$: cycle +11,−12,+22,−21. θ=2 → ($x_{11}$,$x_{12}$,$x_{21}$,$x_{22}$)=(2,1,0,2), cost 11; per-unit change 2−5+1−4=−6.

**Assignment:** $x_{ij}$≥0, each row sum=1, each column sum=1; min$\sum c_{ijx_ij}$. Integral optimum exists; explicit binaries unnecessary for pure assignment. Forbidden pairs remove arcs; unequal set sizes may need dummies. Greedy can fail: costs [2,3;10,20], greedy 2+20=22, optimum 3+10=13.

**Shortest path as flow:** send 1 unit, $b_s=1$,$b_t=-1$, others 0; min$\sum c_{ijf_ij}$. With nonnegative costs an optimal simple path exists. Negative cycles can spoil this formulation. Reliable path with independent arc success $p_{ij}$: maximize product p ⇔ minimize $\sum -\log p$.

**Maximum flow:** max F; out−in=F at source, −F at sink, 0 elsewhere; 0≤f≤u. Or add t→s arc with cost −1 and minimize circulation cost. Any s–t cut capacity bounds flow; max flow=min cut. Residual forward capacity=u−f, reverse=f; augment along residual paths, including reverse arcs to undo earlier choices.

**Time-expanded network:** node=(location,time). Travel arcs move to arrival time; holdover arcs keep inventory at a location. Flow conservation carries stock between periods; required deliveries are lower bounds/fixed flows. Never let a departure use inventory that arrives later.

**Process flexibility:** source→plant capacity C_i; plant→product only if capable; product→sink capacity D_j. Max flow=maximum sales. Dedicated sales=$\sum$min(C_i,D_i); full flexibility=min($\sum$C_i,$\sum$D_j). Example capacities (100,100,100), demand (20,140,140): dedicated 220, full 300.

**Project / crashing:** activity start $s_i$, duration $d_i$, reduction $r_i\in[0,R_i]$. Precedence i→j: $s_j\ge s_i+d_i-r_i$. Min$\sum k_{ir_i}$; finish≤deadline; s≥0. Earliest finish is longest-path length; critical path has zero slack. Example A(3)→C(4), B(2)→C: duration 7; crashing B alone cannot reduce 7. Multiple critical paths may need simultaneous reductions.

## 8 · INTEGRALITY & BOUNDS [L7]

**TU:** every square subdeterminant is 0,+1,−1. With integer RHS/bounds, a TU constraint system has integral BFS. Directed incidence and bipartite assignment matrices are TU. Cramer: x_Bi=det(B with col i replaced by b)/det B. Integer b and det B=±1 ⇒ integer x_B. Merely having entries 0,±1 is insufficient: [1 1;1 −1] has det −2. Extra side constraints can destroy integrality.

**LP relaxation:** drop integrality, retain 0≤x≤1 for binaries. MAX: incumbent≤IP optimum≤LP bound. MIN: LP bound≤IP optimum≤incumbent. Integral LP optimum solves IP; fractional LP may round infeasibly or suboptimally.

**Branch & bound:** solve node LP; prune if infeasible, integer optimum, or bound no better than incumbent. Fractional x_j=v: branch $x_j\le\lfloor v\rfloor$ and $x_j\ge\lceil v\rceil$. MAX global upper bound=max bounds of live nodes (and incumbent); MIN global lower bound=min. Stop when no node can improve incumbent.

**Lecture B&B:** max $1000x_1+950x_2+350x_3$, $100x_1+80x_2+50x_3\le200$, binaries. Root LP=(1,1,.4), UB=2090. $x_3$=0 ⇒ (1,1,0), incumbent=1950. $x_3$=1 ⇒ (.7,1,1), UB=2000; branch $x_1$: 0 gives 1300; 1 gives (1,.625,1), UB=1943.75. Both prune ⇒ optimum 1950.

## 9 · INTEGER MODELLING TOOLKIT [L7]

**Binary logic (all variables 0/1):** at most k: $\sum x\le k$; at least k: $\sum x\ge k$; exactly k: $\sum x=k$. A⇒B: a≤b. A⇒not B: a+b≤1. Not A⇒B: a+b≥1. A iff B: a=b. A⇒(B AND C): a≤b,a≤c. A⇒(B OR C): a≤b+c. (A AND B)⇒C: a+b−1≤c.

**Exact AND:** z=ab ⇔ z≤a,z≤b,z≥a+b−1. For n variables: $z\le x_i\ \forall i$, $z\ge\sum x_i-n+1$. **Exact OR:** $z\ge x_i\ \forall i$, $z\le\sum x_i$. Implication alone needs only its direction: "if any x=1 then z=1" needs $z\ge x_i$; "if all x=0 then z=0" needs $z\le\sum x_i$.

**Threshold:** if $\sum x_i\ge k$ then z=1: $\sum x_i\le(k-1)+(n-k+1)z$ (1≤k≤n). For "iff", also $\sum x_i\le kz$. Example n=8,k=5: $\sum x\le4+4z$; if sum=5, z=0 is impossible.

**On/off:** 0≤q≤Uy means no production when y=0. If production is either 0 or at least L: Ly≤q≤Uy. Fixed cost: objective K y+cq. q≤Uy alone does not force y=0 when q=0; a positive fixed cost does so at optimum if no other constraint forces y=1.

**Binary×continuous:** z=yq, y binary, 0≤q≤U: 0≤z≤Uy; z≤q; z≥q−U(1−y). Example q=7,U=10: y=0 forces z=0; y=1 forces z=7. Do not put yq directly in an MILP.

**Conditional rows:** y=1⇒$a^Tx\le b$: $a^Tx\le b+M(1-y)$. y=1⇒$a^Tx\ge b$: $a^Tx\ge b-M(1-y)$. Equality needs both. "Either $a^Tx\le b$ OR $c^Tx\le d$": first ≤b+My, second ≤d+M(1−y). Choose M from valid bounds, as small as possible; test y=0 and y=1.

**Knapsack:** max$\sum v_ix_i$; $\sum w_ix_i\le W$; x binary. Value/weight greedy solves fractional knapsack, not generally 0–1. Lecture counterexample: values (1000,950), weights (100,80), W=100. LP=(.2,1), z=1150; round to (0,1), z=950; IP optimum (1,0), z=1000.

**Facility location:** open $y_i$ binary, shipments $q_{ij}\ge0$; min$\sum F_iy_i+\sum c_{ijq_ij}$. Demand $\sum_i q_{ij}=d_j$; link $q_{ij}\le d_j\,y_i$; capacity $\sum_j q_{ij}\le C_iy_i$ if capacitated. Uncapacitated aggregate bound $\sum_j q_{ij}\le(\sum d_j)y_i$. Without a linking constraint, a closed facility can ship for free.

**Lot sizing:** end inventory $I_t=I_{t-1}+q_t-d_t$; $I_t\ge0$ (no backlog), $I_0$ fixed; $0\le q_t\le M_t\,y_t$, $y_t$ binary. Min$\sum(K_ty_t+c_tq_t+h_tI_t)$. Set terminal inventory if specified; $M_t=\sum_{\tau=t}^{T} d_\tau$ is valid when excess production is unnecessary. Airplane example: d=(3,2,3,2), K=2, h=.2; q=(10,0,0,0), I=(7,5,2,0), total=2+.2(14)=4.8 million.

## 10 · ROUTING, PRICING & SCHEDULING

**Directed TSP:** $x_{ij}$ binary, i≠j; min$\sum c_{ijx_ij}$. One outgoing $\sum_{j\ne i}x_{ij}=1$ and one incoming $\sum_{i\ne j}x_{ij}=1$. These alone form a **cycle cover**, not necessarily a tour; assignment optimum is a lower bound for min TSP.

**DFJ subtour cuts:** $\sum_{i,j\in S,i\ne j}x_{ij}\le|S|-1$ for all proper S⊂N with 2≤|S|≤n−1. Example subtour 2→3→2 violates $x_{23}+x_{32}\le1$. Equivalently, every proper nonempty subset has ≥1 outgoing arc. Many constraints; can add violated cuts iteratively.

**MTZ alternative:** depot=1; $1\le u_i\le n-1$ for i=2,…,n; $u_j\ge u_i+1-(n-1)(1-x_{ij})$, i≠j, i,j≥2. $u_i$ records visit order. Exclude depot from these comparisons so return is possible; u can be continuous. Summing selected order constraints around a customer-only cycle gives 0≥cycle length, impossible.

**Pickup before drop:** u_drop≥u_pick+1. Open route: allow free artificial return to start (cost 0), or use a separate terminal node. Forbid unavailable arcs instead of using literal infinity as a solver coefficient.

**Arrival-time formulation:** start $t_1$=0; $t_j\ge t_i+service_i+travel_{ij}-M_{ij}(1-x_{ij})$ for j≠1. Windows $a_j\le t_j\le b_j$. $M_{ij}\ge b_i+service_i+travel_{ij}-a_j$ (use valid depot bounds). No time-propagation row back to fixed $t_1$; use a separate return-time variable if needed. Positive travel times eliminate customer subtours; zero-time cycles may still need cuts.

**Mini time example:** $t_1$=0, $travel_{12}$=3, $travel_{23}$=4. Selected 1→2→3 implies $t_2\ge3$,$t_3\ge7$. Deadline $b_3$=6 rules it out. **Total route time** $\sum c_{ijx_ij}$ differs from **total customer waiting** $\sum_{j\ne1}t_j$; choose the requested objective.

**Job shop:** operation start $s_{ik}\ge0$; within-job order k→ℓ: $s_{iℓ}\ge s_{ik}+p_{ik}$. Shared machine k for jobs i,j; y=1 means i before j: $s_{jk}\ge s_{ik}+p_{ik}-M(1-y)$; $s_{ik}\ge s_{jk}+p_{jk}-My$. Completion $C_i\ge s_{ik}+p_{ik}\ \forall k$. Min$\sum C_i$ for total completion/waiting (fixed release times and processing durations); min $C_{max}$ with $C_{max}\ge C_i$ for makespan. These objectives are different.

**Tiny schedule:** same machine, durations A=2,B=3, both released at 0. A→B: completion (2,5), sum 7; B→A: (5,3), sum 8. Makespan 5 in both. Precedence alone does not prevent two jobs sharing a machine at once.

**Multiproduct price grid:** price $p_m$; $x_{nm}$=1 iff product n uses m. Demand D(n,m,k) depends on own price and **total** price k. For each possible fixed k, max$\sum p_mD(n,m,k)x_{nm}$; $\sum_m x_{nm}=1\ \forall n$; $\sum_{nm}p_mx_{nm}=k$. Solve all k; choose highest value. Equally spaced M prices, N products ⇒ N(M−1)+1 totals vs $M^N$ price vectors.

**Sudoku:** $x_{ijk}$=1 iff cell (i,j) contains digit k. Exactly one digit/cell; each digit once in each row, column and 3×3 block; fix clues $x_{ij,given}$=1. Feasibility objective min 0.

## 11 · DYNAMIC PROGRAMMING [L8]

**Five things to write:** stage, state, feasible action, transition, one-stage cost/reward; then value function, boundary and Bellman equation. State must retain all past information needed for future decisions; policy maps (stage,state) to action. Save subproblem values instead of recomputing. Store argmin/argmax to reconstruct decisions.

**Convention here:** t=number of periods **remaining** (matches lecture). Deterministic min-cost: $V_t(s)=\min_{a\in A_t(s)}\{c_t(s,a)+V_{t-1}(f_t(s,a))\}$; $V_0(s)$=terminal cost. Reward maximization: replace min/cost by max/reward. Infeasible states get +∞ for min, −∞ for max; do not give them value 0.

**Principle of optimality:** after a chosen action, remaining actions must be optimal for the reached state. Compute backward from boundary; follow policy forward. DP works when subproblems share sufficient states; large state spaces cause the curse of dimensionality. Fibonacci memoization/tabulation takes O(n), unlike repeated recursion.

**Shortest path on staged DAG:** $V_0$(destination)=0; $V_k(i)=\min_{j:(i,j)\in A}\{c_{ij}+V_{k-1}(j)\}$. Unreachable states +∞. At each state compare complete "edge cost + future value", not just shortest outgoing edge.

**Lecture path example:** V(H)=23,V(I)=29. V(E)=min(16+23,18+29)=39; V(F)=min(18+23,11+29)=40; V(G)=min(15+23,13+29)=38. V(B)=55,V(C)=56,V(D)=59. V(A)=min(22+55,19+56,15+59)=74. Optimal route A→D→F→I→J: 15+19+11+29=74.

**0–1 knapsack DP:** F(i,b)=best value using first i items with budget b. F(0,b)=0 (b≥0). If $w_i\le b$: $F(i,b)=\max\{F(i-1,b),v_i+F(i-1,b-w_i)\}$; else F(i,b)=F(i−1,b). Take vs skip; no fractional item. O(nW) with integer weights/budget (pseudo-polynomial). For 1D memory update b **downward** to avoid reusing an item.

**Lecture knapsack:** w=(2,3,4,6), v=(8,12,15,19), W=10. After items 1–3, best at budget 10 is 35 (items 1+2+3, weight 9). Compare item 4: 19+F(3,4)=19+15=34 ⇒ skip. Optimum 35, unused budget 1. Full budget need not be used.

**Small DP cells:** F(2,5)=max(F(1,5)=8,12+F(1,2)=20)=20. F(3,6)=max(20,15+F(2,2)=23)=23. Backtrack: if "take" wins, subtract item weight and move to previous item.

## 12 · STOCHASTIC DP & INFORMATION

**Bellman:** $V_t(s)=\min_{a\in A_t(s)} E_D[c_t(s,a,D)+V_{t-1}(f_t(s,a,D))]$. Choose action before D realizes; average over D, then optimize over actions. Do not replace E[V(next state)] by V(E[next state]). For demand-dependent feasible actions, respect what is known when choosing.

**Order matters:** E[min_a cost(a,D)] ≤ min_a E[cost(a,D)]; the left allows knowing D before choosing a (perfect information), so is too optimistic for a pre-demand decision. For rewards, inequality reverses. Adaptive policies use only observations already available.

**Inventory recurrence (lost sales):** state s=stock before ordering, order q≥0, sales=min(s+q,D), next stock=$(s+q-D)^+$. $V_t(s)=\min_q\{cq+E[-p\cdot\min(s+q,D)+h(s+q-D)^+ +V_{t-1}((s+q-D)^+)]\}$. If salvage v/unit: $V_0(s)=-vs$. If no salvage: $V_0(s)$=0. Specify order/storage limits if present.

**Lecture newsvendor:** c=.4,p=1; independent daily D∈{0,1} equally likely; zero holding/salvage; inventory carries over. One day: $V_1(0)=\min(0,.4-.5)=-.1$ (order 1); $V_1(1)=V_1(2)=-.5$ (order 0). Existing stock is already paid for.

**Two-day decision from stock 0:** q=0: $0+V_1(0)=-.1$. q=1: $.4-.5+½V_1(0)+½V_1(1)=.4-.5-.05-.25=-.4$. q=2: $.8-.5+V_1(1)=-.2$. Choose $q_1$=1, expected profit .4.

**Policy:** day 2 order 1 if no stock remains ($D_1$=1); order 0 if stock remains ($D_1$=0). Expected sales=1, expected purchases=1.5 ⇒ profit 1−.4(1.5)=.4. Best open-loop fixed orders (1,0): one unit sells with probability ¾ ⇒ profit .75−.4=.35. Value of adaptation=.05. Open-loop commits all actions beforehand; feedback uses observed state.

**Optional single-period shortcut:** price p, cost c, salvage v with p>c>v: order a demand quantile at critical ratio (p−c)/(p−v). Discrete demand: smallest q with F_D(q) at least this ratio. In lecture example ratio=.6; F(0)=.5<.6, F(1)=1 ⇒ q*=1.

## 13 · OPTIMAL STOPPING: SECRETARY

**Assumptions:** n candidates in uniformly random order, only relative ranks observed, accept/reject immediately, no recall; objective probability of selecting overall best. Any fixed interview position succeeds with probability 1/n.

**State r:** current candidate is a record/best-so-far (r=1) or not (r=0); index k also matters. Let W_k(r) be best success probability at candidate k. Boundary W_n(0)=0,W_n(1)=1. Reject/continue value R_k=[W_(k+1)(1)+kW_(k+1)(0)]/(k+1). Thus W_k(0)=R_k; W_k(1)=max{k/n,R_k}. Accepting a non-record has success 0; accepting a record succeeds with probability k/n.

**Worked n=4:** k=4: $(W_4(0),W_4(1))=(0,1)$. k=3: $R_3=¼$; $W_3=(¼,¾)$. k=2: $R_2=(¾+2\times¼)/3=5/12$; $W_2=(5/12,½)$. k=1: $R_1=(½+5/12)/2=11/24$; $W_1(1)=11/24>¼$, so reject candidate 1. Then accept first record among candidates 2–4; success 11/24≈45.8% vs fixed-position 25%.

**Large n rule:** observe/reject about n/e≈.368n candidates, then accept first later record. If none appears, selecting the last candidate cannot recover success. Threshold policy exploits information; it does not guarantee choosing the best.

**Final recall:** MIN simplex r≥0; MAX simplex r≤0. MAX primal ≤ ↔ MIN dual ≥. Network out−in=supply. MAX LP bound is upper; MIN LP bound is lower. DP needs boundary + feasible actions + transitions + expectation timing. Always give a numerical check with your model.

# DBA5103 • OPERATIONS RESEARCH & ANALYTICS

**SIDE 3 / 4 · COURSE SUMMARY: WHAT EACH LECTURE COVERED (L1–L8)**

## L1 · INTRODUCTION: WHAT OR IS

**In one line:** pick decisions to maximize/minimize an objective subject to constraints, then choose the right tool to solve the model.

**Covered:** course road map LP → network flow → IP → DP. Software: Python + Gurobi, Excel Solver. Application areas: manufacturing, transport, airline/hotel/retail. Cases: **Ford flexible manufacturing** (dedicated vs fully flexible plants; evaluate a design with LP, max flow or simulation); **Uber/Grab** dispatch (greedy matching vs assignment/shortest path), surge pricing (supply–demand model: LP/linear algebra if non-profit, nonlinear if profit-maximizing), **UberPOOL** as a TSP with pickup before drop-off (LP, IP, heuristics); **Rue La La** price optimization (IP/LP); markdown/dynamic pricing (LP, DP).

**Numbers to know:** capacities (100,100,100), demands (20,140,140): dedicated sells 220, full flexibility 300. Dispatch times [2,3;10,20]: greedy 2+20=22 min, optimal 3+10=13.

**Quiz angle:** say which model type fits a story (LP / network / IP / DP) and why.

**Watch out:** greedy, one-step-at-a-time choices can be far from optimal.

## L2 · LINEAR PROGRAMMING & REFORMULATION

**In one line:** write a word problem as a linear model, then rewrite it in a standard form that theory and solvers use.

**Covered: Wyndor Glass** prototype: max $3x_1+5x_2$; $x_1 \le 4$, $2x_2 \le 12$, $3x_1+2x_2 \le 18$, $x \ge 0$; solved by the graphical/contour method, optimum (2,6), z=36. Linear algebra review: vectors, matrices, transpose, products, identity, inverse, linear combinations, when A is invertible. General LP with ≥, ≤, = rows → standard form. Tricks: free variables, absolute values, min–max. Regression with L1 / L∞ / L2 error and outlier sensitivity. Sparse recovery (compressed sensing, image reconstruction): min $\sum|x_j|$ s.t. Ax=b.

**Formulas:** min $c^Tx$, Ax=b, x≥0. ≤ row: +slack; ≥ row: −surplus. Free $x=x^+-x^-$. $|r| \le t \Leftrightarrow -t \le r \le t$. L1 and L∞ fits are LPs; L2 is quadratic.

**Quiz angle:** formulate from words; convert to standard form; linearize an absolute-value objective.

**Watch out:** "linear" means linear in the decision variables (no xy, $x^2$). The $t \ge \pm r$ trick needs the objective to push t down.

## L3 · GEOMETRY & THE SIMPLEX METHOD

**In one line:** an LP optimum sits at a corner (BFS); simplex moves from corner to a better adjacent corner until no move helps.

**Covered:** Wyndor in Excel Solver and Gurobi; keep model and data separate. 2-D geometry: feasible region = polyhedron, extreme points, four outcomes (unique, multiple, unbounded, infeasible). Binding constraints and slack. BFS in n dimensions = n linearly independent active constraints. Basic solutions in standard form: pick m columns, set the rest to 0, solve, check ≥0. Degeneracy. Convex sets/functions (local optimum = global). Adjacent BFSs. Simplex: find a BFS → reduced costs → entering variable → ratio test → pivot.

**Formulas:** $x_B=B^{-1}b$; $r_j=c_j-c_BB^{-1}A_j$. MIN optimal when all $r_j \ge 0$. Example max $x_1+x_2$; $x_1+2x_2 \le 3$, $2x_1+x_2 \le 3$ → (1,1), z=2.

**Quiz angle:** list basic solutions and mark the feasible ones; do one pivot; name binding constraints.

**Watch out:** θ=0 is degeneracy, not infeasibility; the ratio test uses only decreasing variables; an unbounded region ≠ an unbounded objective.

## L4 · DUALITY & SENSITIVITY

**In one line:** every LP has a dual that gives bounds and resource prices; sensitivity asks how the answer moves when data change.

**Covered:** Lagrangian relaxation (min $x^2+y^2$, x+y=1: $g(p)=p-p^2/2$, best p=1, z=½); LP dual from the Lagrangian; dual of the dual = primal; weak and strong duality; economic view (a market buys Wyndor plant hours at y=(0,1.5,1), value 36 = primal); complementary slackness; what-ifs: add a constraint, add a variable, change a RHS (shadow price + range), change a profit coefficient; Excel and Gurobi sensitivity reports.

**Numbers to know:** plant 3 +1 hour → z=37; +9 hours → z=42, **not** 45 (outside range). Product 1 profit 3→10: old plan alone gains ≥14; new corner (4,3), z=55 (+19).

**Quiz angle:** write a dual; use complementary slackness to get y*; read a shadow price and its valid range.

**Watch out:** shadow prices hold only inside the range and one change at a time; binding does not mean a positive price.

## L5 · LP CASES: MARKDOWN & ROBUST SOURCING

**In one line:** two real LP cases, plus planning for the worst case when data are uncertain.

**Covered: markdown management** for seasonal goods (stock fixed before the season; only price can change); data → demand model → optimize; LP in "effective weeks" at each price; effect of initial stock; dynamic adjustment (re-solve with remaining stock and time). **Dual-sourcing drug case:** nominal LP; uncertain raw-material yields; ad-hoc fix; "prepare for the worst" robust LP; box and budgeted uncertainty; duality turns the worst case into linear rows.

**Numbers to know:** markdown optimum 11⅔ weeks at $60 + 3⅓ at $54, revenue $116,750. Drug: nominal profit ≈8,819.66 can miss the ingredient target; robust ≈8,294.57 and switches to raw 1.

**Quiz angle:** formulate the markdown LP; write a robust row (pick the worst coefficient); contrast sensitivity vs uncertainty.

**Watch out:** purchase cost is sunk; with q≥0 in a ≥ row, the worst case uses the **lowest** yields.

## L6 · NETWORK FLOW MODELS

**In one line:** many LPs are flows on a graph; that structure gives integer answers and fast special methods.

**Covered:** graph terms (directed/undirected, path, cycle, acyclic, connected, tree, spanning tree with n−1 arcs, bipartite, complete); min-cost flow (out − in = supply); node–arc incidence matrix; special cases: transportation, assignment, shortest path, max flow; initial solutions by northwest corner or least cost; m+n−1 basic cells form a spanning tree; transportation simplex with potentials. Applications: container shipping between Singapore, Ho Chi Minh and Bangkok (time-expanded network); expediting and outsourcing for a complex machine (project crashing); process flexibility with random demand.

**Formulas:** integer supplies/capacities ⇒ integer optimal flow (TU). Max flow = min cut. Complete graph n(n−1)/2 edges.

**Quiz angle:** write flow-balance rows; build an initial transport BFS; name the special model.

**Watch out:** need ∑b=0; one balance row is redundant; basic arcs can carry zero flow.

## L7 · INTEGER PROGRAMMING

**In one line:** binary and integer variables model yes/no choices and logic; solve with LP relaxation + branch and bound.

**Covered:** MIP, pure IP, BIP; **if-then logic for project selection (Side 4)**; product of binaries; binary × continuous; capital budgeting; facility location; flexibility design with K demand samples; TSP (brute force, subtour cuts, MTZ); UberPOOL pickup/drop-off routing; Sudoku; Rue La La pricing (fix total price k); job scheduling; LP relaxation; Cramer's rule and totally unimodular matrices; branch and bound.

**Numbers to know:** knapsack LP 1150, rounded 950, IP 1000. B&B: root 2090 → optimum 1950.

**Quiz angle:** translate if-then rules; write big-M rows; run a few B&B nodes; explain why rounding fails.

**Watch out:** assignment rows alone allow subtours; use the smallest valid M.

## L8 · DYNAMIC PROGRAMMING

**In one line:** split a sequential problem into stages, summarize the past in a state, and solve backward with the Bellman equation.

**Covered:** deterministic vs stochastic DP; Fibonacci (save sub-results, don't recompute); value function, Bellman equation, boundary; staged shortest path A→J; choosing the state; investment/knapsack with budget as state; Go (~250 moves per state, ~150 per game: far too many states); stochastic two-day newsvendor; open-loop vs adaptive policies; secretary (optimal stopping) with a threshold.

**Numbers to know:** path A→D→F→I→J = 74; knapsack 35; newsvendor profit .4 adaptive vs .35 open-loop; secretary n=4 success 11/24; large n skip ≈37%.

**Quiz angle:** define stage/state/action/transition; fill a DP table; write Bellman with E[·].

**Watch out:** take the expectation **before** the min when you act before demand is seen; infeasible states get ±∞.

# DBA5103 • OPERATIONS RESEARCH & ANALYTICS

**SIDE 4 / 4 · TURNING "IF… THEN…" RULES INTO MATH (BINARY LOGIC)**

**Set-up:** six candidate projects; $x_i$=1 means "do project i", $x_i$=0 means "skip it". Every rule below becomes a **linear** constraint on these switches, so the model stays an IP the solver can handle. **Good habit:** test each rule by plugging in 0s and 1s and checking it accepts every allowed combo and rejects every forbidden one.

## 14 · THE CORE RULES (projects $x_1…x_6$, each a 0/1 switch)

| Rule in words | Math | Why it works | 0/1 test: the only combos it blocks |
|---|---|---|---|
| At most one of projects 1, 2, 3 | $x_1 + x_2 + x_3 \le 1$ | Only one switch can be on | any two or all three on |
| Exactly two of projects 2–6 | $x_2 + x_3 + x_4 + x_5 + x_6 = 2$ | Exactly two switches on | any count other than 2 (0, 1, 3, 4, 5 on) |
| If 1, then 4 | $x_1 \le x_4$ | If $x_1 = 1$, $x_4$ is forced to be 1 | $(x_1, x_4)$ = (1, 0) only |
| If 2, then NOT 5 | $x_2 + x_5 \le 1$ | They can't both be on | $(x_2, x_5)$ = (1, 1) only |
| If not 1, then 6 | $x_1 + x_6 \ge 1$ | At least one of them must be on | $(x_1, x_6)$ = (0, 0) only |
| 4 and 5 go together | $x_4 = x_5$ | Both on or both off | (1, 0) and (0, 1) |
| If 3, then 1 or 5 | $x_3 \le x_1 + x_5$ | Doing 3 needs at least one helper | $(x_3, x_1, x_5)$ = (1, 0, 0) only |
| If 3 AND 4, then 6 | $x_6 \ge x_3 + x_4 - 1$ | The right side only reaches 1 when both are on | $(x_3, x_4, x_6)$ = (1, 1, 0) only |

## 15 · MORE RULES IN THE SAME STYLE

| Rule in words | Math | Why it works / what it blocks |
|---|---|---|
| If 2, then 3 AND 4 | $x_2 \le x_3$ and $x_2 \le x_4$ | One row per required project. $2x_2 \le x_3 + x_4$ is also valid for 0/1 but gives a weaker LP relaxation |
| If 1 or 2, then 3 | $x_1 \le x_3$ and $x_2 \le x_3$ | Either trigger alone forces 3. Not $x_1 + x_2 \le x_3$ (that wrongly bans doing 1 and 2 together) |
| If 1 and 2, then NOT 3 | $x_1 + x_2 + x_3 \le 2$ | Blocks only all three on |
| If 1, then at least 2 of {3, 4, 5} | $x_3 + x_4 + x_5 \ge 2x_1$ | $x_1 = 0$ gives ≥ 0 (no effect); $x_1 = 1$ gives ≥ 2 |
| If 1, then none of {2, 3} | $x_2 + x_3 \le 2(1 - x_1)$ | $x_1 = 1$ makes the right side 0; $x_1 = 0$ makes it 2 (no effect) |
| Exactly one of 1, 2 (either/or) | $x_1 + x_2 = 1$ | Blocks both off and both on |
| 1 and 2 can't both be skipped | $x_1 + x_2 \ge 1$ | Same as "if not 1 then 2" and "1 unless 2" |
| 1 only if 4 / 1 if 4 | $x_1 \le x_4$ / $x_4 \le x_1$ | "Only if" points at the requirement: "1 only if 4" = "if 1 then 4" |
| $x_1 = x_2 \cdot x_3$ (all binary) | $x_1 \le x_2$, $x_1 \le x_3$, $x_1 \ge x_2 + x_3 - 1$ | $x_1 = 1$ exactly when both are on (an AND switch) |
| $x_1 = x_2 \cdot x_3$ ($x_2$ binary, $0 \le x_3 \le M$) | $-Mx_2 \le x_1 \le Mx_2$; $-M(1 - x_2) \le x_1 - x_3 \le M(1 - x_2)$ | $x_2 = 0$ forces $x_1 = 0$; $x_2 = 1$ forces $x_1 = x_3$ |
| If y = 1, then $a^Tx \le b$ (x continuous) | $a^Tx \le b + M(1 - y)$ | y = 1: the real row; y = 0: M makes it slack. M = valid upper bound on $a^Tx - b$ |
| Either $a^Tx \le b$ OR $c^Tx \le d$ | $a^Tx \le b + My$; $c^Tx \le d + M(1 - y)$ | y picks which row must hold; the other is switched off |
| Produce x > 0 only if facility y is open | $0 \le x \le My$ | y = 0 forces x = 0; add fixed cost K·y to the objective |

## 16 · METHOD, CHECKS & TRAPS

### A RECIPE THAT WORKS FOR ANY RULE

**1. Write it as logic:** IF (conditions) THEN (results).

**2. Find the one bad case:** the rule is broken only when **every** condition is true and the result is false. Your constraint must cut off exactly that corner.

**3. Literals:** "i is done" → $x_i$; "i is skipped" → $(1 - x_i)$.

**4. One-line formula for "THEN A OR B":** ∑(1 − condition literals) + ∑(result literals) ≥ 1. For "THEN A AND B", write one row per result.

**Examples:** If 3 AND 4 then 6: $(1-x_3)+(1-x_4)+x_6 \ge 1 \Rightarrow \mathbf{x_6 \ge x_3+x_4-1}$. If 3 then 1 or 5: $(1-x_3)+x_1+x_5 \ge 1 \Rightarrow \mathbf{x_3 \le x_1+x_5}$. If not 1 then 6: $x_1+x_6 \ge 1$. If 2 then not 5: $(1-x_2)+(1-x_5) \ge 1 \Rightarrow \mathbf{x_2+x_5 \le 1}$.

### THE 0/1 TEST, WORKED

List every 0/1 combo (2 variables = 4 rows, 3 = 8). Mark what the words allow, then plug in.

$\mathbf{x_6 \ge x_3+x_4-1}$, combos $(x_3,x_4,x_6)$: 000 ✓ 001 ✓ 010 ✓ 011 ✓ 100 ✓ 101 ✓ **110** ✗ 111 ✓. Only "3 and 4 on, 6 off" fails. Correct.

$\mathbf{x_3 \le x_1+x_5}$, combos $(x_3,x_1,x_5)$: only **100** gives 1 ≤ 0 ✗; all 7 others pass. Correct.

**Wrong try $\mathbf{x_6 \ge x_3}$ and $\mathbf{x_6 \ge x_4}$:** blocks 100-style rows (3 on, 4 off, 6 off) the words allow. That means "if 3 OR 4 then 6": too strong.

### COMMON TRAPS

**Direction:** "if 1 then 4" is $x_1 \le x_4$. Writing $x_4 \le x_1$ means the converse "if 4 then 1".

**Contrapositive is the same rule:** "if not 4 then not 1" also gives $x_1 \le x_4$.

**Three look-alikes:** "if not 1 then 6": $x_1+x_6 \ge 1$. "If 1 then not 6": $x_1+x_6 \le 1$. "Exactly one of 1, 6": $x_1+x_6 = 1$.

**OR uses a sum:** "if 3 then 1 or 5" is $x_3 \le x_1+x_5$. $2x_3 \le x_1+x_5$ would demand both.

**AND in the condition needs −1:** $x_6 \ge x_3+x_4-1$; when only one is on the right side is 0 and nothing is forced.

**An if-then never forces anything when the condition is off.** Check the "condition off" rows are all still allowed.

**Big-M:** choose the smallest valid M; test y = 0 and y = 1 separately.

### QUICK DRILL (cover the answers)

**a)** At least 2 of projects 1–4 → $x_1+x_2+x_3+x_4 \ge 2$

**b)** If 5, then neither 1 nor 6 → $x_5+x_1 \le 1$, $x_5+x_6 \le 1$

**c)** If not 4, then 5 and 6 → $1-x_4 \le x_5$, $1-x_4 \le x_6$

**d)** If 2, then exactly one of 3, 4 → $x_3+x_4 \ge x_2$, $x_3+x_4 \le 2-x_2$

**e)** At most 3 of projects 1–6, and if 6 then 1 → $\sum x_i \le 3$, $x_6 \le x_1$

**f)** Do 1 if and only if both 2 and 3 → $x_1 \le x_2$, $x_1 \le x_3$, $x_1 \ge x_2+x_3-1$